

N7.40

Дано:

$$N = \frac{N_0}{2}$$

Найти:

T

Решение:

$$N \cdot \frac{3}{2} \mu T = \frac{3}{2} \mu (T - dT) (N - dN) + 2 \mu dN T$$

$$3 \cancel{N} T = 3 \cancel{N} T - 3 dT N - 3 dN T + 4 dN T$$

$$dN T = 3 dT N$$

$$\frac{dN}{N} = 3 \frac{dT}{T} \Rightarrow$$

$$\ln \frac{N}{N_0} = 3 \ln \frac{T}{T_0}$$

$$\left(\frac{N}{N_0} \right)^{\frac{1}{3}} = \frac{T}{T_0} = \left(\frac{1}{2} \right)^{\frac{1}{3}} \Rightarrow T = T_0 \left(\frac{1}{2} \right)^{\frac{1}{3}} = 217 (K)$$

N8.15

Дано:

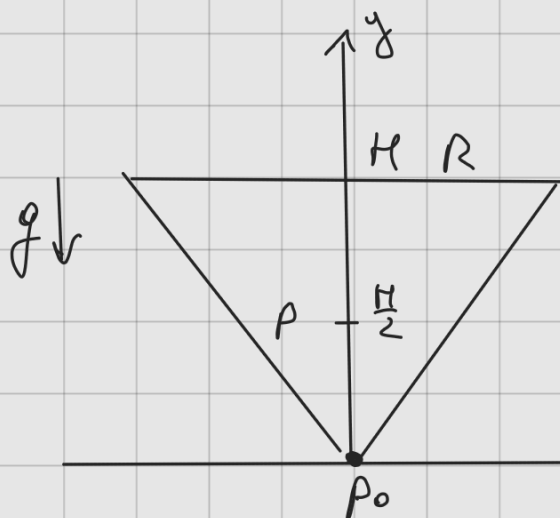
$$H, g, m$$

Найти: T

Решение:

$$p = p_0 e^{-\frac{p - p_0}{\kappa T}}$$

$$pV = \nu R T$$



$$1) n = n_0 \exp\left(-\frac{mgh}{\kappa T}\right)$$

$$2) V = S \cdot dy; S = \pi \cdot \left(\frac{R \cdot y}{H}\right)^2$$

$$dV = \frac{\pi R^2}{H^2} \cdot y^2 dy$$

$$dN = n \cdot dV$$

$$3) dN = n_0 \frac{\pi R^2 y^2 dy}{H^2} \exp\left(-\frac{mgy}{kT}\right)$$

$$\left(\frac{\pi R^2 y^2 \exp\left(-\frac{mgy}{kT}\right)}{H^2}\right)'_y = \frac{2y \pi R^2 \exp(-)}{H^2} - \frac{\pi R^2 y^2}{H^2} \cdot \exp(-) \cdot \frac{mg}{kT} = 0$$

$$2 - y \cdot \frac{mg}{kT} = 0 \Rightarrow T = \frac{y \cdot mg}{2k} ; y = \frac{H}{2}$$

$$T = \frac{mgH}{4k}$$

№8.28

Дано:

$$G = 100 \text{ g}$$

$$\alpha = \frac{dn}{n dr} = 10 \frac{1}{\text{cm}}$$

$$\rho = 1,12 \text{ g/cm}^3$$

$$\rho_0 = 0,92 \text{ g/cm}^3$$

$$T = 293 \text{ K}$$

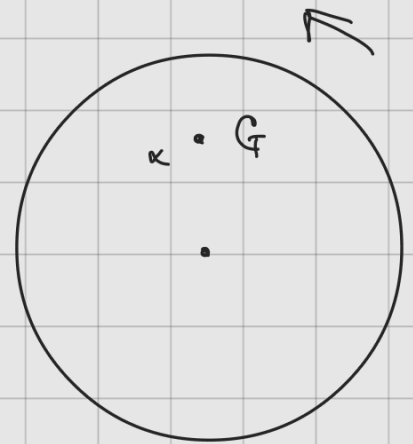
Найти: M

Решение:

$$1) p = p_0 \exp\left(-\frac{\pi - \pi_0}{kT}\right)$$

$$n = n_0 \exp\left(-\frac{\pi}{kT}\right)$$

$$n = n_0 \exp\left(\frac{mgR}{kT}\right)$$



$$\frac{dn}{n dr} = \frac{mg}{kT} ;$$

$$\frac{dn}{n dr} = \frac{Mg}{RT} \cdot \frac{p - p_0}{p} = \alpha \Rightarrow$$

$$\Rightarrow M = \frac{\rho \alpha RT}{g}$$

$$G(p-p_0)$$